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Do Language Models Know When They're Hallucinating References?

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Abstract

State-of-the-art language models (LMs) are famous for "hallucinating" references. These fabricated article and book titles lead to harms, obstacles to their use, and public backlash. While other types of LM hallucinations are also important, we propose hallucinated references as the "drosophila" of research on hallucination in large language models (LLMs), as they are particularly easy to study. We show that simple search engine queries reliably identify such hallucinations, which facilitates evaluation. To begin to dissect the nature of hallucinated LM references, we attempt to classify them using black-box queries to the same LM, without consulting any external resources. Consistency checks done with *direct* queries about whether the generated reference title is real (inspired by Kadavath et al. [10], Lin et al. [12], Manakul et al. [13]) are compared to consistency checks with *indirect* queries which ask for ancillary details such as the authors of the work. These consistency checks are found to be partially reliable indicators of whether or not the reference is a hallucination. In particular, we find that LMs often hallucinate *differing* authors of hallucinated references when queried in independent sessions, while *consistently* identify authors of real references. This suggests that the hallucination may be more a generation issue than inherent to current training techniques or representation.

1 Introduction

Language models (LMs) famously hallucinate[1], meaning that they fabricate strings of plausible but unfounded text. As LMs become more accurate, their fabrications become more believable and therefore more problematic. A primary example is "hallucinated references" to non-existent articles with titles readily fabricated by the LM. For instance, a real New York Times article entitled "When A.I. Chatbots Hallucinate" leads with a ChatGPT[2]-fabricated New York Times article titled "Machines Will Be Capable of Learning, Solving Problems, Scientists Predict" [24]. In this work, we study the problem of hallucinated references.

The hallucinated reference problem is worth study for multiple reasons. First, as we discuss, hallucinated references can easily be evaluated and debunked. Second, hallucinated references impact applications, as LMs help generate literature reviews [11] for the exploration and citation of related work and may assist in writing of paper reviews [15]. Third, due to the deployment of these models, the problem of hallucinated references has pushed beyond an academic curiosity to attract the attention of masses [e.g., 4, 24, 14, 22, 18] and has been highlighted as a problem in the medical domain

[1]Though it is an anthropomorphism, we use the term *hallucinate* due to its widespread adoption, following the use-theory of meaning [25]. We use the terms *hallucinate* and *fabricate* interchangeably.

[2]https://openai.com/blog/chatgpt

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[4, 1] where hallucinations could be extremely harmful. Finally, the insights gained from studying hallucinated references may apply to hallucination in domains beyond references.

A motivating question for this work is, *why do LMs hallucinate, and what can be done about it?* Is it a problem of LM *representation*, a problem of *training* (maximizing next-word likelihood), or a problem due to the way they are used for *generation*? Specifically, we investigate whether an LM itself can be used to detect whether or not an output it has produced is a hallucination, without any external resources. While this does not provide a complete answer to the questions of why and what to do, it does inform the discussion. In particular, to the extent that LMs can be used to detect their own hallucinations, this suggests that the hallucination problem is not inherently one of training or representation but is rather one of generation because the models contain enough information to at least reduce the hallucination rate.

In this work, by hallucinations we are referring to fabricated text that has *little or no grounding in the training data*. Note that this has been referred to as *open-domain hallucination* to distinguish it from *closed-domain hallucination* [see, e.g., 9], which is often studied in summarization and machine translation, where the fabrications are defined relative to a specific source document to be summarized or translated (as opposed to the training data). The two types of hallucinations are different in terms of what is often considered a hallucination: background information based on the training corpus is often defined to be a hallucination in the study of closed-domain hallucinations (assuming it is not in the source document, e.g., the text to be translated). However, open-domain hallucination has attracted significant recent attention within scientific communities and journalism. In this work, when we refer to hallucinations we are referring to absolute (i.e., open-domain) hallucinations.

Groundedness versus correctness The opposite of fabrication is *groundedness* in the sense of being based on the training corpus rather than accuracy in the sense of being a true fact (a genuine publication, in the case of references). We define hallucination to be fabricated text, meaning text that is not grounded in this training set. In contrast, correctness is evaluated with respect to ground-truth answers. This distinction is called *honesty* versus *truthfulness* by Evans et al. [6]. For example, the common misconception that "people use 10% of their brains" is grounded because it is almost surely mentioned in the training data, either exactly or in various paraphrased versions. However, it is not scientifically correct. Much work on hallucination conflates groundedness and accuracy, often equating hallucination with fallacy and evaluating hallucinations using accuracy on fact-based assessments, without regard to the training data [9]. We adopt the groundedness definition of hallucination even though it may often be less clear-cut and more difficult to evaluate than factuality.

Evaluating groundedness Perfectly evaluating hallucinations would require access to the LM's training data. An advantage of the hallucinated reference problem is ease of (approximate) evaluation in that exact-match Web search is a reasonable heuristic for groundedness. This is because the vast majority of article titles present in the training data are included in Web search results—articles are meant to be published and shared, and publishers aim to make their work discoverable by search. Furthermore, references generally have titles that are specific enough not to spuriously occur on the Web. Regarding other types of hallucinations, besides article names, which cannot be as easily evaluated, we still hope that our methodology and findings would apply, even if evaluating those types of hallucinations would require access to the training data.

Direct queries Our work builds upon and is inspired by two recent works that show how to use black-box generative LMs to assess confidence in generations, without consulting external references or inspecting weights. In particular, Kadavath et al. [10] introduce multiple direct black-box strategies for using an LM to extract confidence estimates by querying the language models on question-answer problems. Manakul et al. [13] apply a similar direct self-consistency check called SelfCheckGPT to identify relative hallucinations in a summarization context. These queries are direct true/false correctness queries. We test similar approaches in the context of hallucinated references. Black-box generative approaches stand in contrast to the work that either introspects the weights on LMs [2] or that consults existing databases [7].

Indirect queries In addition, we suggest a new approach using what we call *indirect queries*. A direct query may ask, *Is the following paper real?* while an indirect query may ask, *Who are the authors of this paper?*, as illustrated in Fig. 1. Answers are then generated to the indirect query in $i > 1$ independent sessions, and tested for consistency. The motivation for indirect queries comes from investigative interviews, where detectives are advised to interview individuals separately and ask open-ended questions. For instance, consistency may be better evaluated by asking multiple witnesses

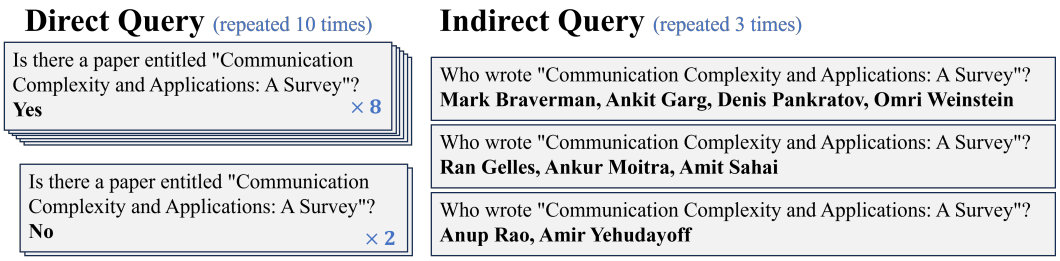

Figure 1: Example direct vs. indirect LM queries for predicting whether a given paper title is hallucinated or grounded. Direct queries are binary, repeated multiple times to estimate a probability. Indirect queries are open-ended, and their answers are compared to one another, using the LM, to output an agreement fraction. Language model generations are indicated in **boldface**. Prompts in this figure have been shortened for illustrative purposes.

to *"Describe in detail what the suspect was holding"* rather than asking, *"Was the suspect holding a gun in their right hand?"* [23]. In the context of reference hallucination, our hypothesis is that the likelihood of multiple generations agreeing on the same authors for a hallucinated reference would be smaller than the likelihood of multiple responses to a direct query indicating that the reference exists.

Contributions There are several contributions of this work. First, we perform a systematic LM study of hallucinated references, enabling us to compare hallucination rates across LMs. Second, we introduce indirect queries for evaluating hallucinations. Third, we compare these to direct queries, inspired by studies in LM question-answering [10] and summarization-based hallucinations [13]. A conclusion of our work for reducing hallucination is the recognition that changing the generation pipeline can certainly help, while it is less clear if training or representation changes are necessary.

2 Related Work

Open-domain hallucination were discussed in the context of GPT-4 [16, 3], due to their prevalence and potential danger, Bubeck et al. [3, page 82] write:

> *Open domain hallucinations pose more difficult challenges, per requiring more extensive research, including searches and information gathering outside of the session.*

We show that open domain hallucinations can in fact be addressed, at least in part, without consulting external resources.

As mentioned, there are multiple definitions of hallucination. In this work, we use the term hallucinations to mean fabricated text that is not grounded in the training data. Factually incorrect generations can be decomposed into two types of errors: grounded errors which may be due to fallacies in the training data (e.g., that people use only 10% of their brains) and ungrounded errors. These two types of errors may need different techniques for remedy. The grounded errors may be reduced by curating a training set with fewer errors or other techniques such as RLHF [17]. However, the ungrounded errors which we study[3] are a fascinating curiosity which still challenge the AI community and one which is not clearly addressable by improving training data. The distinction is further elucidated by Evans et al. [6].

There is comparatively little prior work studying *open-domain groundedness* like ours. Some work [e.g., 8] in attribution aims to understand which training examples are most influential in a given output. In recent independent work in the health space, Athaluri et al. [1] did an empirical evaluation of hallucinated references within the medical domain. Similar to our approach, they used a Google search for exact string match as a heuristic for evaluating hallucinations. Our study of hallucinated references enables us to estimate the hallucination rates of different models, and, as discussed in prior work, the hallucination problem interestingly becomes more pressing as models become more accurate because users trust them more [16].

[3] One can also imagine ungrounded correct generations, such as a generated paper title that exists but is not in the training data, but we find these to be quite rare.